



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: X
SUBJECT: PHYSICS

FULL MARKS: 40
DATE: 19.05.2023

General Instructions:

- This paper consists of three printed pages.
- All questions are compulsory.
- Marks will be deducted for spelling errors.

SECTION A

Question 1

Choose the correct answers to the questions from the given options. [1 × 7 = 7]

- (i) Two bodies of masses m_1 and m_2 have equal linear momentum. The ratio of their kinetic energies K_1 and K_2 will be

a) $m_2:m_1$ b) $m_1:m_2$ c) 1:1 d) $m_2^2: m_1^2$

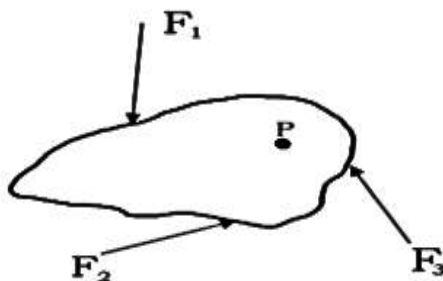
- (ii) Choose the correct option wrt the direction of centripetal force.

a) Towards the centre of the circular path.
b) Away from the centre of the circular path.
c) Normal to the radius at a point where the body is present on the circular path.
d) Direction of this force keeps on changing alternately towards and away from the centre.

- (iii) eV/s is the unit of

a) force b) energy c) power d) mass

- (iv) In the figure shown below, three forces F_1 , F_2 and F_3 are seen acting on a body pivoted at P. If each force acts separately and moment of force is same in each case, then



- a) Force F_1 has the maximum value.
b) Force F_2 has the maximum value.
c) Force F_3 has the maximum value.
d) All forces have same value.

- (v) A body is acted upon by two equal and opposite forces along same lines of action of force. The body will have:
- only rotatory motion
 - only translatory motion
 - both (a) and (b)
 - neither (a) nor (b)
- (vi) The energy change during photosynthesis in plants is
- heat to chemical.
 - light to chemical.
 - chemical to light.
 - chemical to heat.
- (vii) The incorrect statement is:
- A body can be balanced at its centre of gravity
 - The total weight of a body acts at its centre of gravity
 - The algebraic sum of moments of the weights of all the particles constituting a body is zero about its centre of gravity.
 - The centre of gravity of a body depends on how it is placed.

Question 2

- (i) a) What is meant by the term 'moment of force'?
 b) If moment of force is assigned a negative sign what will be the turning tendency of the body?
 c) How can moment of force be made zero despite of application of a non-zero finite force? [3]
- (ii) Mention the conditions for work done to be
- maximum
 - negative
- [2]
- (iii) Why are passengers travelling in a double decker bus allowed to stand in a lower deck, but not in the upper deck? [2]
- (iv) A uniform metre rule of weight 10 gf is pivoted at its 0 mark.
- What is the moment of force that depresses the rule?
 - How can it be made horizontal by applying a least force? [2]
- (v) The power of a motor is 40 kW. At what speed can it raise a load of mass 2000 kg? $[g = 10 \text{ m/s}^2]$ [2]
- (vi) A force is applied on a pivoted body. Mention the ways of changing the direction of rotation of the body. [2]

SECTION B

(Attempt all questions.)

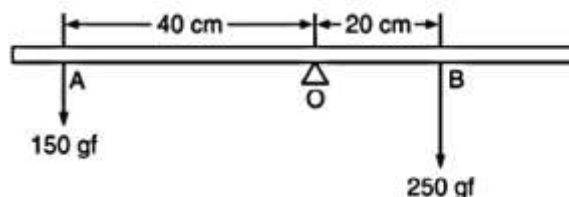
Question 3

[3 + 3 + 4 = 10]

- A force of 100 N acting on a body of mass 20 kg accelerates it uniformly for 10s. If the initial velocity of the body is 5 m/s. Find the work done by the force in 10s, using the concept of work energy theorem.
- Show graphically the variation of different forms of mechanical energy with height, in motion under gravity.
- What is a couple force?
 - Write the expression for moment of couple explaining the term mentioned in the expression.
 - Why does the temperature of water at the bottom of a waterfall differ from the temperature at the top?

Question 4**[3 + 3 + 4 = 10]**

- i) The diagram shows a uniform metre rule weighing 100 gf, placed horizontally on a fulcrum at its centre O. Two weights 150 gf and 250 gf hang from the metre rule as shown in the figure.



- a) Will the rule be in equilibrium?
b) Justify your answer showing relevant calculation.
- ii) A body of mass 0.5 kg falls from a height of 5 m. How much mechanical energy does it possess at any instant? What will be the speed of the object just before it touches the ground? (take $g = 10 \text{ m/s}^2$)
- iii) A stone of mass 'm' is rotated in a circular path with a uniform speed by tying a strong string with the help of your hand. Answer the following questions:
- a) Is the stone moving with a uniform or variable speed?
b) Is the stone moving with a uniform acceleration? In which direction does the acceleration act?
c) What is the work done at any instant during the motion mentioned above?